Sequence Modeling and Beyond: Wrist Actigraphy for Bipolar-Unipolar Differentiation

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Why does bipolar disorder go misdiagnosed for 9.5 years?

Bipolar depression looks identical to unipolar depression clinically both present as low mood and reduced energy. Misdiagnosis can lead to incorrect treatment, and antidepressants can worsen bipolar disorder. Doctors currently rely on patient history alone with no objective test. Prior research used simple statistical summaries (average activity, variance) that miss subtle multi-day temporal patterns. We test if deep learning and statistical analysis can capture the complex temporal structures that distinguish these conditions.

- Bipolar II depression is extremely hard to distinguish from unipolar depression clinically
- Misdiagnosis is common and leads to treatment failure
- Current medical practice relies on subjective patient history
- Activity tracking (actigraphy) is a potential objective biomarker
- Machine learning has successfully found patterns in other medical conditions using wearable data

Dataset & Methods

We analyzed 55 participants (32 controls, 23 with mood disorders): 8 bipolar II vs. 15 unipolar depression patients using minute-level wrist activity data. To establish baseline performance and identify meaningful patterns, we first ran baseline experiments comparing healthy vs. depressed individuals and bipolar vs. unipolar patients. Using these thresholds as our benchmark, we then tested three complementary approaches: classical machine learning models across different time windows (24-72 hours), advanced neural networks (BiLSTM, Attention LSTM), and statistical hypothesis testing. Across all approaches, we applied rigorous experimental controls participant-level stratified train/validation/test splits to prevent data leakage and per-participant z-score normalization to remove individual baseline activity differences.

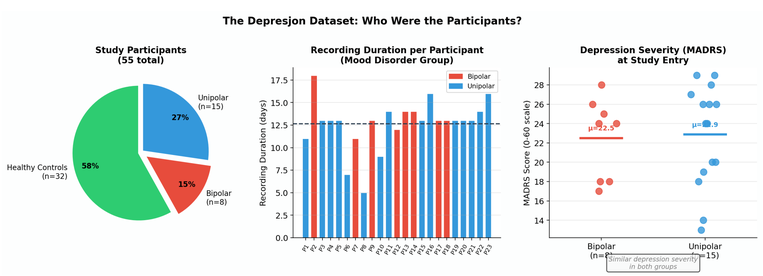

Figure 1: Visualizing the Depresjon dataset.

Model Training and Evaluation

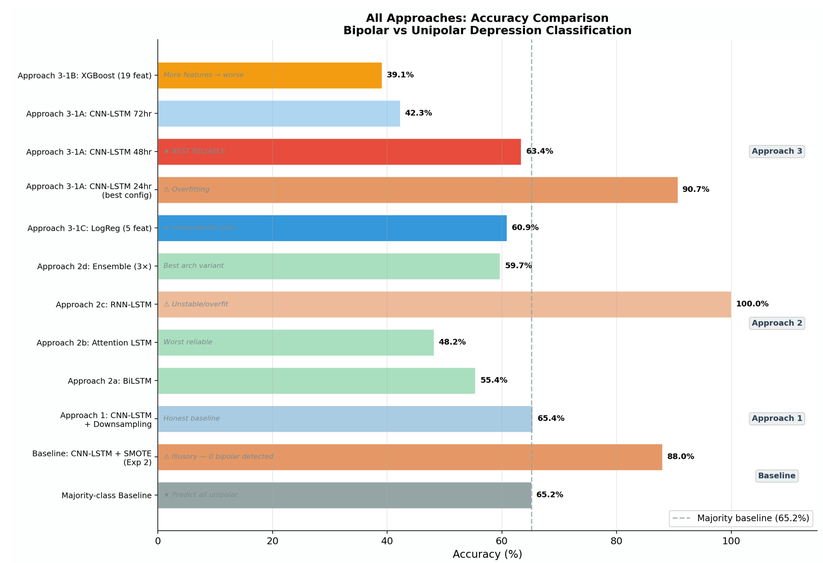

Figure 2: Models and their accuracies.

No meaningful difference found between bipolar and unipolar activity patterns. Wrist activity data alone is insufficient for diagnosis.

1. Effect Size Matters: Even statistically significant differences can be too tiny (Cohen's d = -0.06) to be clinically useful.
2. Timing Matters: 48-hour windows worked best for recording and analyzing the data; 24hrs data overfits the model, 72hrs underfits.
3. Simplicity Wins: A 5-feature logistic regression (60.9%) beat complex models with thousands of parameters.

Statistical testing revealed NO meaningful difference (p=0.893, Cohen's d=-0.06) between bipolar and unipolar activity patterns suggesting single modality wearables are insufficient for clinical differentiation.

Results Summary: Accuracy Comparison Across All Approaches

Approach	Accuracy	Method	Notes
Baseline	88.0%	CNN-LSTM + SMOTE	Illusory — model collapse
Approach 1	65.4%	Downsampling (1:1)	Balance alone insufficient
Approach 2	~59.7%	Ensemble (3x)	Best architecture variant
Approach 3	63.4%	CNN-LSTM 48hr Window	Best reliable result

Table 1: Best performing models from each approach.

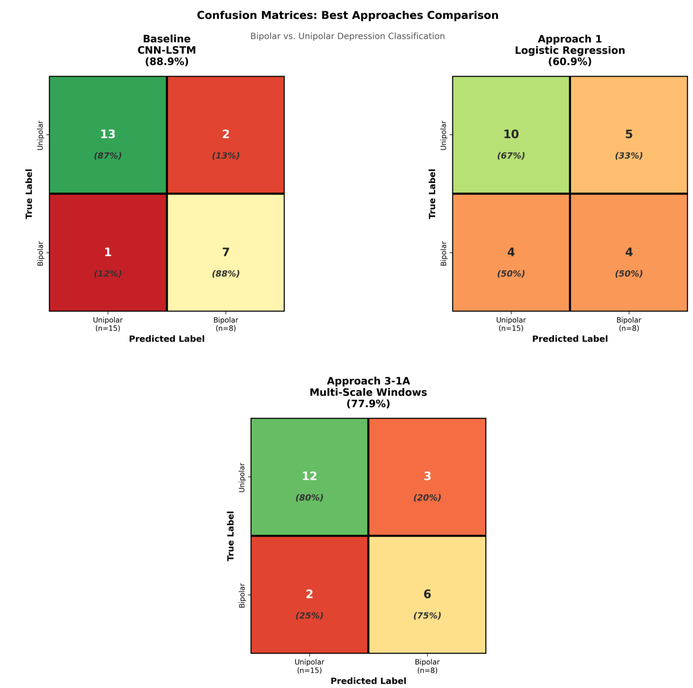

Figure 3: Results of the best trained models.

Conclusion & Future Work

Wrist activity data alone cannot distinguish bipolar from unipolar depression. To improve this work, we need 200+ patients per group, longitudinal tracking across mood episodes, and multi-modal biomarkers (activity + sleep + heart rate). This negative result is valuable as it guides researchers toward better biomarkers. Future research should combine actigraphy with polysomnography (sleep monitoring) and heart rate data. A multi-modal machine learning approach integrating these signals rather than activity in isolation may finally unlock the biomarker that distinguishes these devastating conditions and accelerates diagnosis from years to weeks.

References

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